

- 5 (a) (i) cannot predict when a particular nucleus will decay [1]
or
cannot predict which nucleus will decay next
- (ii) (decay is) not affected by external (environmental) factors [1]
- (iii) fluctuations in (measured) count rate [1]
- (b) (i) Any three points, 1 mark each: [3]
 - large nuclei undergo fission whereas small nuclei undergo fusion
 - fission involves one nucleus splitting into two (or more) (smaller) nuclei
 - fusion involves two nuclei joining together to form one (larger) nucleus
 - fission is (usually) initiated by neutron bombardment
 - fusion is (usually) initiated by (very) high temperatures
- (ii) binding energy per nucleon is greatest at nucleon numbers around 56 (may be shown on sketch graph with axes labelled 'binding energy per nucleon' and 'nucleon number') [1]
both fusion and fission involve an increase in binding energy (per nucleon) [1]
- (c) (i) ${}^{66}_{29}\text{Cu}$ [1]
- (ii) the energy of the decay is fixed / constant [1]
the energies of the beta particles have a (continuous) range of values / varies / not constant [1]
another particle / an (anti)neutrino must possess the extra / remaining energy (difference between energy of the decay and the β kinetic energy) since total momentum has to be conserved [1]
- (iii) Sharp peak at max KE, zero for all other energies (regardless of height) [1]

